

113A Introduction to Computer and Computer Science

Homework Assignment #2

Due: 9/30 12:00:00

In this homework assignment, you will practice the basic concepts of input, branching, conditional, and iterations. As you have learned all the related knowledge during the lectures (if, while, for, exhaustive search, and so on...).

Problem #1: Find the roots

Please write a python program to find the roots of the following equation.

$$x^2 + x - 72 = 0$$

The estimation error of your program should be less than 10^{-4} . The definition of estimation error, E , is

$$E = |\text{your estimation} - \text{roots}|$$

Here is the sample code,

```
print("Problem 1: Find the roots")

# 自己想

# Print the result
print("Find the approximated roots: {:.05f}, {:.05f}".format(r1, r2))
# How many steps do we have?
print("# of guesses = {}".format(numGuesses))

print("="*60)

print("End of this homework.")
```

Attention!

Find the roots.

Importing any additional module or package is strictly prohibited in this homework.

Use this formula if you want to calculate $f'(x)$.

$$f'(x) = \frac{f(x + dx) - f(x)}{dx}$$

Hand in procedure:

As we had mentioned in the lecture, you should list all your collaborators in your programs. Here is the template:

```
"""  
Created on Wed Aug 7 01:23:45 2024  
  
@author: Xi Winnie, student ID  
  
@collaborators: William Lai, his student ID  
"""
```

Please save your homework as a “.py” file, where the file name should follow this format:

113A_hw2_**ID**.py

For example,

113A_hw2_0123456789.py

Please note. **We are not going to accept any homework file with wrong file name, wrong file format, or without signature.** Please double check the content of your file.

Once you have accomplished your works, you can submit your homework to the “E3@NYCU” system. There will be a section for uploading your homework.